

Aryan Nehete

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EDUCATION

University of Toronto

Expected April 2028

BASc in Engineering Science – Engineering Math, Stats, Finance Major & AI Minor

Toronto, Canada

- **5x** Dean's Honours List Scholar | **cGPA:** 3.71 | **Avg:** 85%
- Quantitative Research Team Lead at St. George Capital | Member in UofT Engineering Finance Association
- **Relevant Coursework:** Engineering Finance & Economics, Financial Principles II, Financial Optimization Models, Economic Analysis & Decision Making, Financial Engineering, Praxis (Engineering Design)

EXPERIENCE

Financial Engineering Analyst

May 2026 – Present

Scotiabank

Toronto, Canada

- Leveraged **multithreading** and dynamic runtime estimation to optimize cross-board regression testing, reducing end-to-end execution time by **86%: 105 min to 15 min**.
- Developed a reusable market data ingestion framework to efficiently parse and structure data for downstream pricing computations, reducing model testing time by **50%**.
- Modernized existing test workflows for cloud execution by developing automation scripts to streamline test inputs and remove unnecessary dependencies, improving computational efficiency and portability.

Teaching Assistant

Sept. 2025 – Apr. 2026

University of Toronto

Toronto, Canada

- Led programming lab practicals for **ESC180: Introduction to Computer Programming** and **ESC190: Computer Algorithms and Data Structures** for 100+ first year Engineering Science students.
- Guide students in learning Python and C fundamentals, including functions, OOP, and problem-solving strategies; and how to analyze the efficiency of these strategies with runtime and memory analysis.

Quantitative Research Team Leader

Sept. 2024 – Apr. 2025

St. George Capital, University of Toronto

Toronto, Canada

- Led a team focused on identifying cyclic arbitrage opportunities in decentralized cryptocurrency exchanges.
- Developed **API integrations** and **in-house data structures** to retrieve and store real-time price and liquidity data.
- Applied graph theory and implementing deep-cycle algorithms to detect profitable trade cycles.

PROJECTS

Prediction Market Trading Bot

December 2025

- Developed a algorithm to **find market inefficiencies** on certain prediction markets by watching for rapid price movements, resulting in guaranteed returns of **0.5-2% over 15-min** periods.
- Created historical datasets by recording live data over five days; then back tested various parameters achieving an ROI of **80%+ over five days**.

NBA Prop Predictor

September 2025

- Built an NBA odds service that ingests historical data and serves probabilities via a **FastAPI** web API & **Streamlit** UI.
- Developed **matchup-aware** models for player/team markets with features for opponent defense, pace, home/away, rest, minutes, and recent form; applied probability **calibration** and surfaced clear "why" explanations in-app.
- Validated with rolling backtests over **3 seasons**: **Brier** = 0.089, **calibration error** = 2-3%, **ROC AUC** = 0.78.

Handwriting to LaTeX [Report]

May – Aug. 2025

- Engineered a **CNN (ResNet-18) encoder + LSTM/Transformer decoders** for Handwritten Mathematical Expression Recognition (HMER), converting rasterized InkML strokes into LaTeX strings.
- Optimized training with **PyTorch**, label smoothing, teacher forcing, AdamW, and beam search (k=3-5), achieving **BLEU** $\approx$ 0.71 on **CROHME** test data for long/nested expressions.

TECHNICAL SKILLS

Programming: Python, C++, C, R, MATLAB, Bash, Java, and HTML/CSS

Frameworks and Libraries: PyTorch, NumPy, GYROBI, Flask, BeautifulSoup, and Seaborn

Data and Analysis: SQL, Pandas, Matplotlib, and Financial Modeling & Optimization

Other: Git, JIRA, Advanced MS Excel, Visual Studio Code, PowerPoint